Grade 5 Natural Sciences and Technology: Processing materials and fair testing

Natural Sciences and Technology · Grade 5 · 30 marks · 60 minutes

Educator marking memo

Suggested answers and marking points. Accept equivalent correct responses appropriate to the grade.

Assessment

Read the stimulus and each question carefully.

> *A Grade 5 CAPS example describes mixing sand, gravel, cement and water to make concrete. Learners will prepare two small concrete samples with different amounts of gravel. They will keep the other materials, moulds, setting time and test method the same. After the samples set, an educator will add identical counters one at a time and record how many each sample supports before it cracks. The class will also record material properties before mixing and after the samples set.*

1.1 (5 marks) Name the four materials mixed in the example and name the processed material they make.

- Names sand. (1 mark)
- Names gravel. (1 mark)
- Names cement. (1 mark)
- Names water. (1 mark)
- Names concrete as the processed material. (1 mark)

1.2 (5 marks) State what property information learners should record before mixing and after the samples set. Give two examples of properties to compare and explain why they record both stages.

- Records properties of the starting materials before mixing. (1 mark)
- Names appearance as one observable property. (1 mark)
- Names texture or another valid observable property as a second property. (1 mark)
- Records a property of the set concrete, such as strength or durability. (1 mark)
- Explains that comparing before and after shows how processing changes material properties. (1 mark)

1.3 (6 marks) In the two-sample test, identify what is changed, what is measured and three conditions that should be kept the same. Explain why this makes the comparison fair.

- Identifies amount of gravel as the changed factor. (1 mark)
- Identifies the number of identical counters supported before cracking as the measured result. (1 mark)
- Keeps the amount of sand the same. (1 mark)
- Keeps the amount of cement and water the same. (1 mark)
- Keeps the mould and setting time the same. (1 mark)
- Explains that changing only the gravel amount helps make the comparison fair. (1 mark)

1.4 (7 marks) Based on the planned test, predict which sample will support more counters, or predict that they will support the same number. Explain why your prediction is reasonable, state what result will count as evidence, and give one reason the result should be checked with repeated trials.

- Gives a clear prediction, including a reasoned prediction of no difference. (1 mark)
- Links the prediction to the changed gravel amount without claiming the result is certain. (1 mark)
- Identifies the number of counters supported before cracking as the result to observe. (1 mark)
- Explains that the same counter type and test method allow comparison. (1 mark)
- Recognises that individual small samples may give unusual results. (1 mark)
- Explains that repeated trials help check whether a result is consistent. (1 mark)
- States that the conclusion should be based on recorded results. (1 mark)

1.5 (7 marks) Give suitable results-table headings, including units where needed, for the gravel test. Then describe one way to keep the comparison reliable and one limitation of using small samples to choose material for a real path.

- Includes a sample or gravel-condition heading. (1 mark)
- Includes the gravel amount with a stated unit or defined measure. (1 mark)
- Includes a trial number. (1 mark)
- Includes the number of counters supported before cracking. (1 mark)
- Keeps other ingredients, mould and setting time the same. (1 mark)
- Repeats the test and records results consistently. (1 mark)
- Notes that small samples cannot by themselves prove how a full path will perform. (1 mark)

Marking checks

Confirm section totals equal the stated total. Apply the grade-appropriate memo style in the Grade Profile and moderate open-ended answers before use.

Source basis: DBE CAPS Natural Sciences and Technology · intermediate_grade_4_6. See the grade receipt for the official PDF URL and SHA-256.

DRAFT — Qualified educator review required before classroom use.